

Domain Query

Optimal Agent Graph

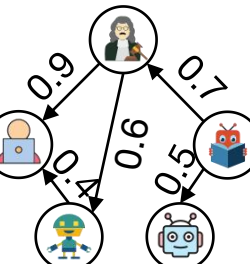
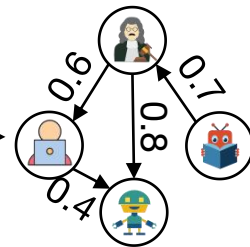
Natural Science

If 3 apples and 4 bananas cost 78 cents, and 4 apples...how many cents does one apple cost?
(A) 8 (B) 10 (C) 12 (D) 14



Humanities History

In which year did the French Revolution begin?
(A) 1776 (B) 1789 (C) 1804 (D) 1815



Query-Conditioned Latent Adaptation



Gradient Reverse

latent-space adversarial discriminator

Domain 1

Domain 2

Domain N

Latent Adaptation Loss

$$\mathcal{L}_{\text{adapt}} = -\mathbb{E}_{(\mathbb{G}, q, d^k) \sim \mathcal{D}} [d^{k\top} \log f_{\Psi}(z)]$$

Task Loss

$$\mathcal{L}_{\text{task}} = -\mathbb{E}_{z \sim q_{\phi}} \mathbb{E}_{\mathbb{G} \sim p_{\theta}} [\Phi(\mathbb{G}, q, y^k)]$$

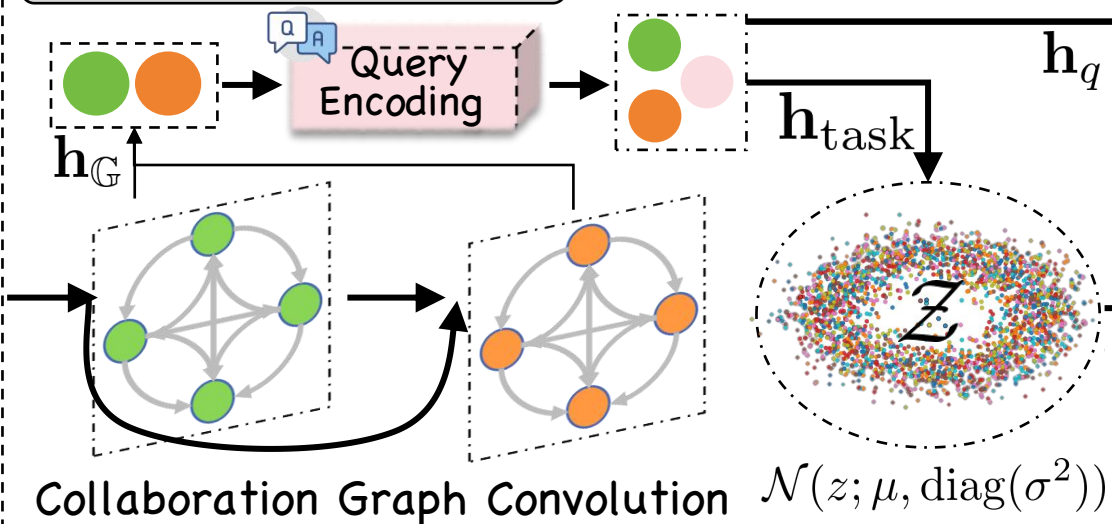
Topology Prior Loss

$$\mathcal{L}_{\text{prior}} = -\mathbb{E}_{z \sim q_{\phi}(z | \mathbb{G}, q)} [\log p_{\theta}(\mathbb{G} | z, q)] + \text{KL}(q_{\phi}(z | \mathbb{G}, q) \| p'_{\theta}(z | q))$$

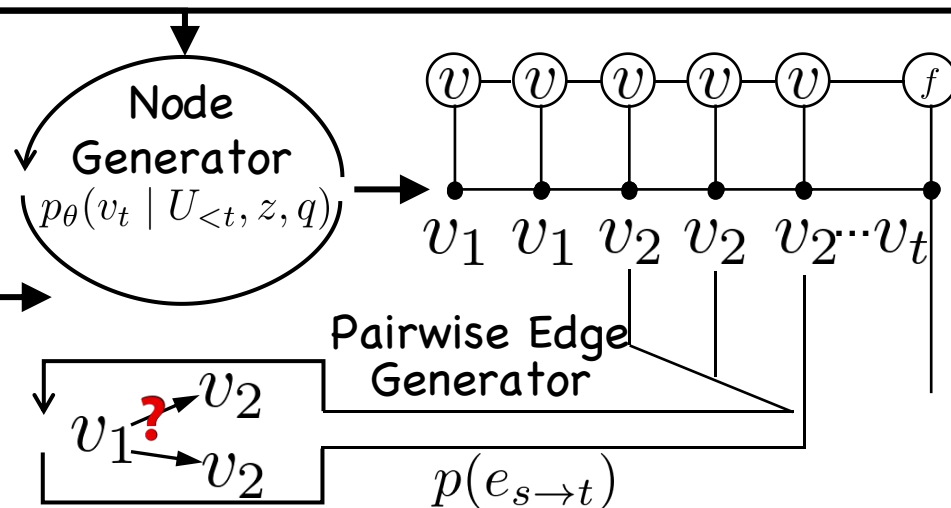
Total Loss

Transferable Topology Prior Learning

① Variational Encoder



② Conditional Generator



③ Conditional Prior

